

Exploratory Data Analysis Report on Chicago Crime Data

1. Executive Summary

This report is based on the notebook and provides a review and extended interpretation of an exploratory data analysis (EDA) conducted on Chicago crime incident data merged from train_2014.csv through train_2024.csv (11 years in total). The notebook's core workflow is: reading yearly CSV files and concatenating them into a unified table (df_train), followed by statistical summaries, feature construction, and visualizations focused on crime-type composition, arrest patterns, temporal regularities (year/month/week/hour), location and spatial distribution (Location Description / latitude–longitude), and correlations/outliers among numeric fields. The data source and field definitions are consistent with the publicly documented “Crimes - 2001 to Present” dataset released via the City of Chicago open data portal, derived from the Chicago Police Department's CLEAR system, de-identified at the block level, and subject to later corrections and classification changes. From the notebook outputs, the 2014–2024 subset has a scale of 2,795,390 rows $\times$ 22 columns, with 33 distinct Primary Type categories. The overall arrest proportion is approximately 18.7% (non-arrested: 81.3%). Among the top-15 crime types, NARCOTICS, WEAPONS VIOLATION, and CRIMINAL TRESPASS show substantially higher arrest rates, while THEFT and DECEPTIVE PRACTICE show comparatively lower arrest rates. These differences suggest structural variation across crime types in terms of “on-scene enforceability / investigability / enforcement contact,” but this report does not make causal claims. Meanwhile, high-frequency Location Description categories concentrate in everyday spaces such as STREET / APARTMENT / RESIDENCE, indicating that prevention and management efforts may benefit from focusing on high-exposure contexts.

2. Data Source and Analytical Scope

The notebook reads locally stored yearly training files (train_2014 to train_2024) and retains 22 fields aligned with the official open-data schema (e.g., ID, Case Number, Date, Block, IUCR, Primary Type, Location Description, Arrest, Domestic, Beat, District, Ward, Community Area, X/Y Coordinate, Latitude/Longitude, etc.). In the official metadata, incidents are extracted from CPD's CLEAR system, de-identified at block level for privacy, and explicitly noted as potentially updated after initial entry.

3. Data Overview and Data Quality

3.1 Scale, Fields, and Types

From the notebook output, the merged df_train has 2,795,390 rows $\times$ 22 columns, with memory usage around 431.9 MB. The data types include: 2 bool columns (Arrest, Domestic), 3 int64 columns (ID, Beat, Year), 7 float64 columns (District, Ward, Community Area, X/Y Coordinate, Latitude, Longitude), and 10 object columns (Case Number, Date, Block, IUCR, Primary Type, etc.).

District/Ward/Community Area appear as floats due to missingness.

3.2 Missing Values and Potential Impact

Missingness is concentrated in geo-related fields: X Coordinate / Y Coordinate /

Latitude / Longitude / Location each has 44,355 missing records (~1.59%), and Location Description has 13,007 missing (~0.47%). Administrative fields have relatively few missing values (District 1, Ward 57, Community Area 200).

Missingness affects:

1. spatial maps/plots after dropna() (systematic exclusion of some incidents), and
2. Top-N counts by Location Description (implicit exclusion of missing-location rows).

4. Temporal Pattern Analysis

4.1. Yearly Trend

The notebook converts Date to datetime and extracts year, then produces a stacked bar chart by year and Primary Type. Visually, total annual incident counts appear relatively high during 2014–2019, show a notable decline around 2020–2021, and rise again during 2022–2024. This pattern may align with multiple external factors, but given the official notice that records and classifications can be updated and the notebook does not include controls, the report treats this as a descriptive observation only (see Figure 1).

4.2. Monthly Trend and Monthly Arrest Rate

By analyzing YearMonth (monthly aggregation) and plots “monthly incident count + monthly arrest rate (dual axis).” shown in Figure 2 , two key patterns are visible:

1. A stable within-year seasonality: many years show higher levels mid-year and lower levels near year-start or year-end, consistent with the notebook’s note that “summer higher, winter lower.”
2. Arrest rate appears higher around 2014–2015 (~0.25–0.32), then declines over time and reaches a lower range around 2021–2022 (~0.10–0.15), with a mild rebound later. Without inferential testing or mechanism decomposition, this report does not claim causal explanations.

4.3. Day-of-Week and Holidays

Arrest proportions fluctuate only slightly (True ~ 18.1%–18.9%)(see Figure 3), suggesting limited weekday effect on arrest share under this aggregation. Holiday True is ~18.3%, non-holiday True ~18.7%, again a small difference. Given that holidays.US() scope is not specified and no statistical testing is applied, this should be interpreted as “weak differences under current implementation.”

5. Spatial Distribution Study

5.1. Top-15 Location Types and Arrest Rates

The notebook visualizes Top-15 Location Description categories (counts) and then computes Arrest True/False stacks and arrest rates for these Top-15 locations (see Figure 4).

5.2 Latitude–Longitude Scatter and Spatial Clustering by Crime Type

After dropna() on geo fields, the notebook overlays scatter plots for THEFT,

HOMICIDE, KIDNAPPING, and CRIMINAL SEXUAL ASSAULT, using a city-range window. The plot indicates that spatial distribution is not uniform and that some types show stronger local clustering.

Important notes for interpretation:

- The plot uses Latitude on the x-axis and Longitude on the y-axis (opposite the common “longitude as x” convention).
- The title states “2020–2024,” but the shown code segment does not clearly demonstrate an explicit year filter. This should be clarified in replication by adding an explicit Year filter and stating axis conventions.

6. Crime Correlation Analysis

The overall pie chart we draw in notebook shows Arrest=True $\approx$ 18.7% and Arrest=False $\approx$ 81.3%, consistent with the official definition of Arrest.

6.1 Arrest Rates for Top-15 Crime Types

The table shown in Figure 5 summarizes incident counts and approximate arrest rates inferred from the notebook’s Top-15 plots.

6.2 Numeric Correlations and Outliers

The notebook computes and visualizes a correlation matrix across numeric fields (excluding is_holiday) shown in Figure 6:

1.X Coordinate is nearly perfectly correlated with Longitude, and Y Coordinate with Latitude, reflecting coordinate-system mapping.

2.Beat is highly correlated with District, consistent with nested administrative structure.

3.ID shows high correlation with year (~ 0.89 in the heatmap), indicating temporal ordering; for predictive modeling, using ID directly can risk time leakage.

Boxplots further suggest extreme outliers in coordinate fields (e.g., near-zero X/Y, lat/long outside typical Chicago range).

Appendix

Figure 1

Number of Crimes by Primary Type and Month-Year

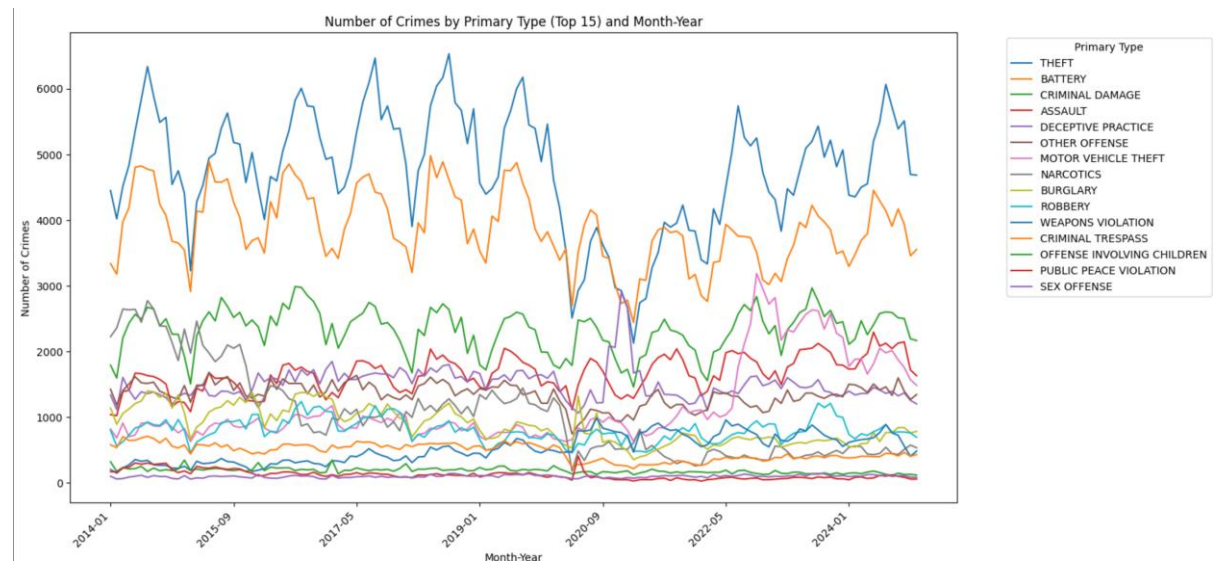


Figure 2

Monthly Crime Trends with Arrest Rate

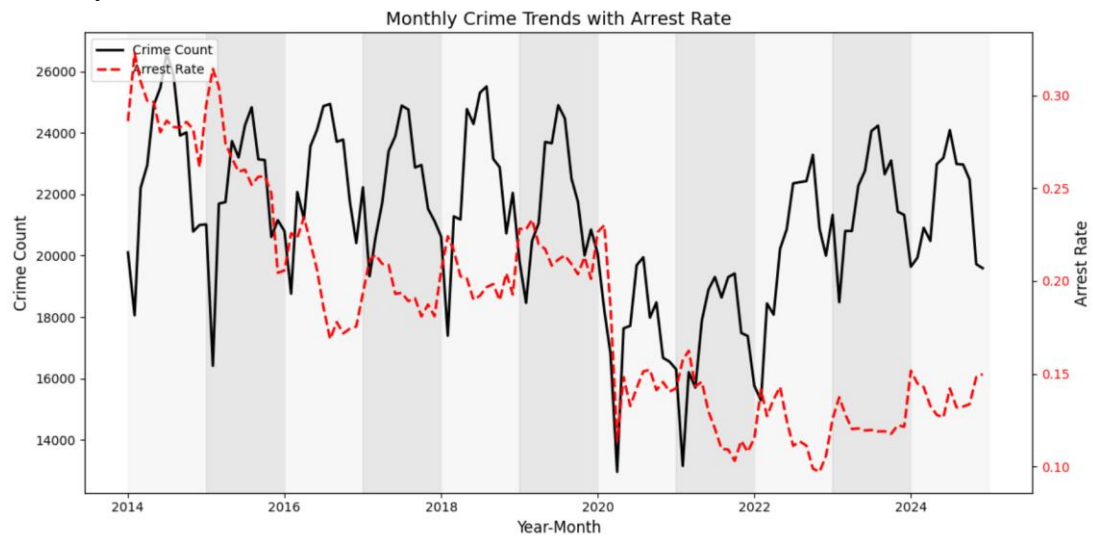


Figure 3
Crime Count by Day of the Week

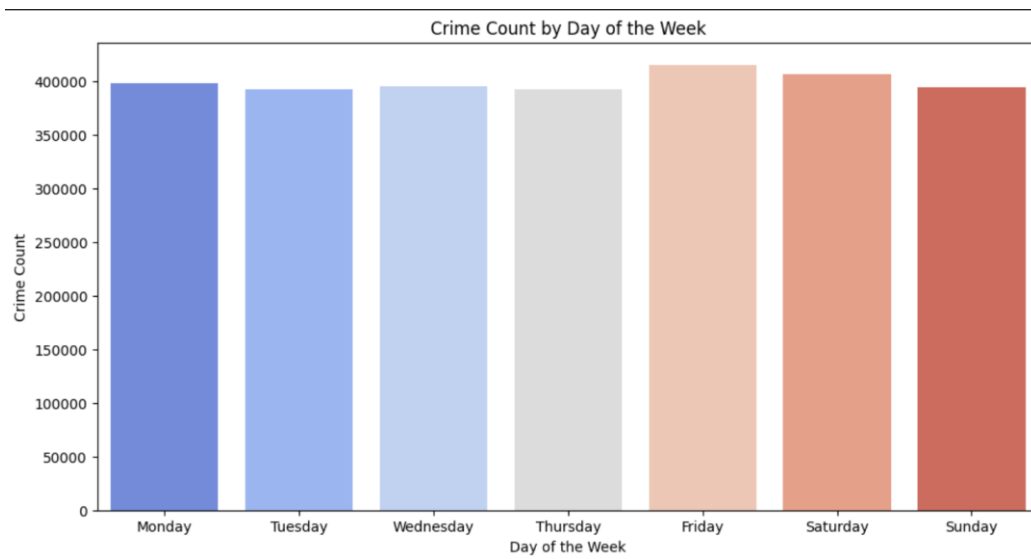


Figure 4
Top 15 Crime Locations

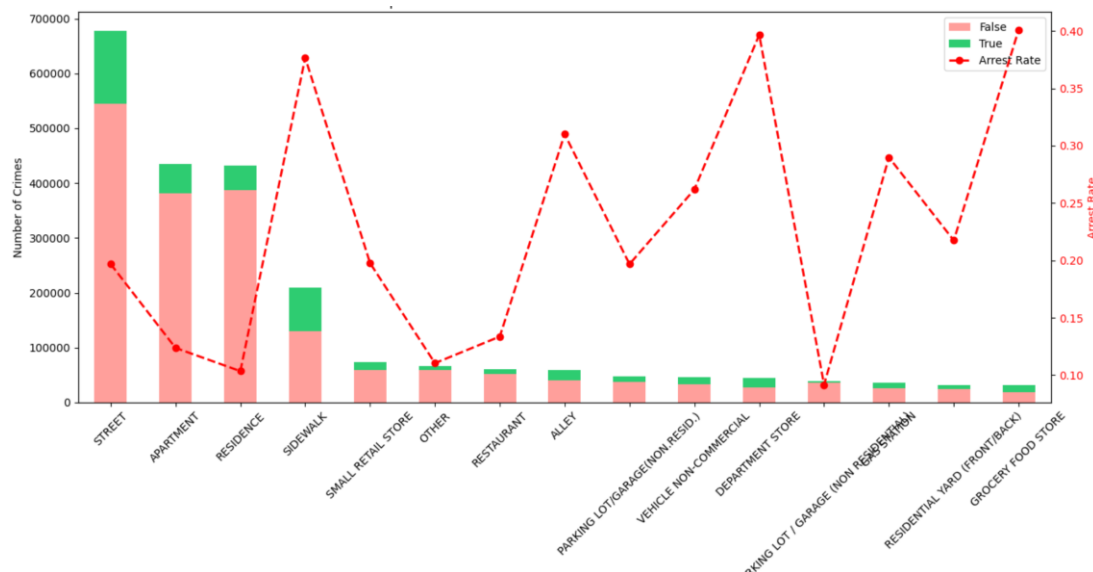


Figure 5
Top-15 Crime Arrest Rates

Primary Type	Incident count	Arrest rate % (approx.)
THEFT	627,835	8.4
BATTERY	510,642	19.4
CRIMINAL DAMAGE	306,921	5.1
ASSAULT	218,560	15.8
DECEPTIVE PRACTICE	198,404	4.2
OTHER OFFENSE	177,292	19.6
MOTOR VEHICLE THEFT	154,614	5.4
NARCOTICS	135,857	100.0
BURGLARY	115,363	5.2
ROBBERY	105,859	8.0
WEAPONS VIOLATION	69,048	66.7
CRIMINAL TRESPASS	62,265	51.0
OFFENSE INVOLVING CHILDREN	24,386	15.0
PUBLIC PEACE VIOLATION	15,750	66.7
SEX OFFENSE	12,833	20.0

Figure 6
Feature Correlations

